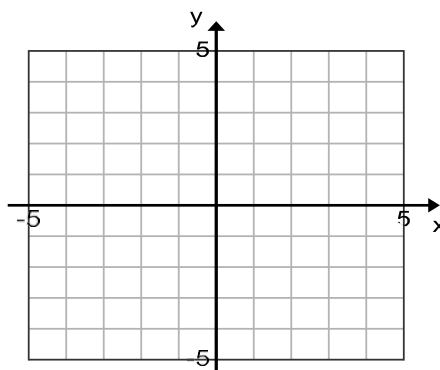
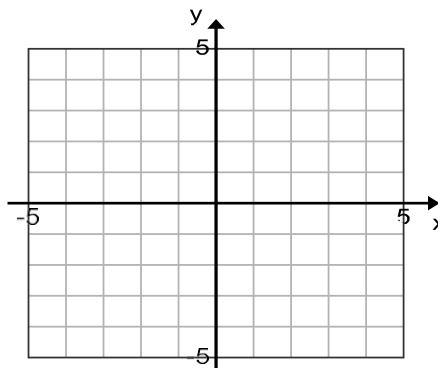

HOMEWORK DAY 14,15,16– *Exponential Function (Section 9.2-9.5)*

1. Graph the function below and determine the domain, range and HA.

(a) $p(x) = 3^{x-4} - 1$



(b) $k(x) = -\left(\frac{1}{3}\right)^x + 2$



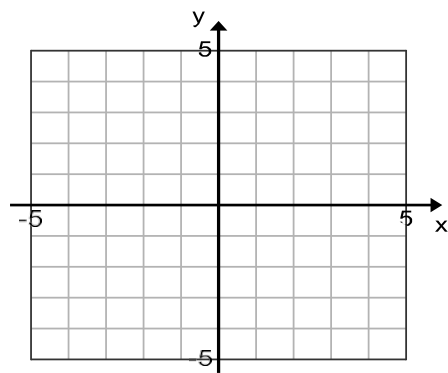
2. Solve for the missing variable.

(a) $2^{-3y+1} = 16$

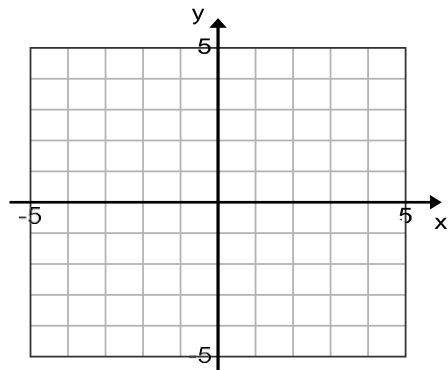
(b) $8^{2x-5} = 32^{x-6}$

3. Graph the function below and determine the domain, range and asymptotes.

(a) $y = \log_3(x - 1) - 3$



(b) $y = -\log_3 x$



4. Convert exponential or logarithmic form.

(a) §9.3 11

(b) §9.3 13

(c) §9.3 15

(d) §9.3 19

(e) §9.3 23

5. §9.5 81

6. Simplify the logarithm.

(a) $\log_3 9 =$

(b) $\log_5 5 =$

(c) $\log 100,000,000 =$

(d) $\log_2 \left(\frac{1}{16}\right) =$

(e) $\log \left(\frac{1}{10}\right) =$

(f) $\ln e^6 =$

(g) $\ln \left(\frac{1}{e^3}\right) =$

(h) $\log_{\frac{1}{7}} 49 =$

(i) $\log_{\frac{1}{2}} \left(\frac{1}{32}\right) =$

(j) $\log 0.00001 =$

(k) $\log_{\frac{3}{2}} \left(\frac{4}{9}\right) =$

(l) $\log_3 \sqrt[5]{3} =$

7. Separate logarithm

(a) §9.4 28

(b) §9.4 30

8. Combine logarithm.

(a) §9.4 55

(b) §9.4 56

9. For the T/F, if false, replace the right side with something that makes the statement TRUE.

(a) $\ln 10 = \frac{1}{\log e}$

(b) $\log_8 \left(\frac{1}{w} \right) = -\log_8 w$

(c) $\log \left(\frac{x}{y} \right) = \frac{\log x}{\log y}$

(d) $\log_4(3d) + \log_4 1 = \log_4(3d)$

10. Solve for x

(a) $8^{2x-3} = \frac{16}{2^{x-2}}$

(b) $e^{x+1} = 2^{3x}$

$$(c) 10 = \frac{5}{1 - 2e^{-0.3t}}$$

$$(d) \log_6(7x - 2) = 1 + \log_6(x + 5)$$

$$(e) 2 \ln \sqrt{x} + \ln(x - 4) = \ln(3x - 10)$$